

Enterprise-Grade Hybrid Fraud Detection System

Technology Stack

Python: Core programming language for backend and data processing.

FastAPI: High-performance asynchronous framework for the REST API.

XGBoost: Advanced gradient boosting library used for the ML behavioral model.

SHAP: Game-theory tool used for Explainable AI (XAI) and model transparency.

PostgreSQL: Primary relational database for storing transactions and features.

SQLAlchemy: Object-Relational Mapper (ORM) for secure database interactions.

Streamlit: Web framework used to build the human-in-the-loop Case Management Dashboard.

Docker & Compose: Containerization tools for isolated, reproducible MLOps deployment.

JSON Logic: Framework used to power the deterministic business Rule Engine.

IMAP/SMTP: Standard email protocols used for asynchronous document verification.